CSC

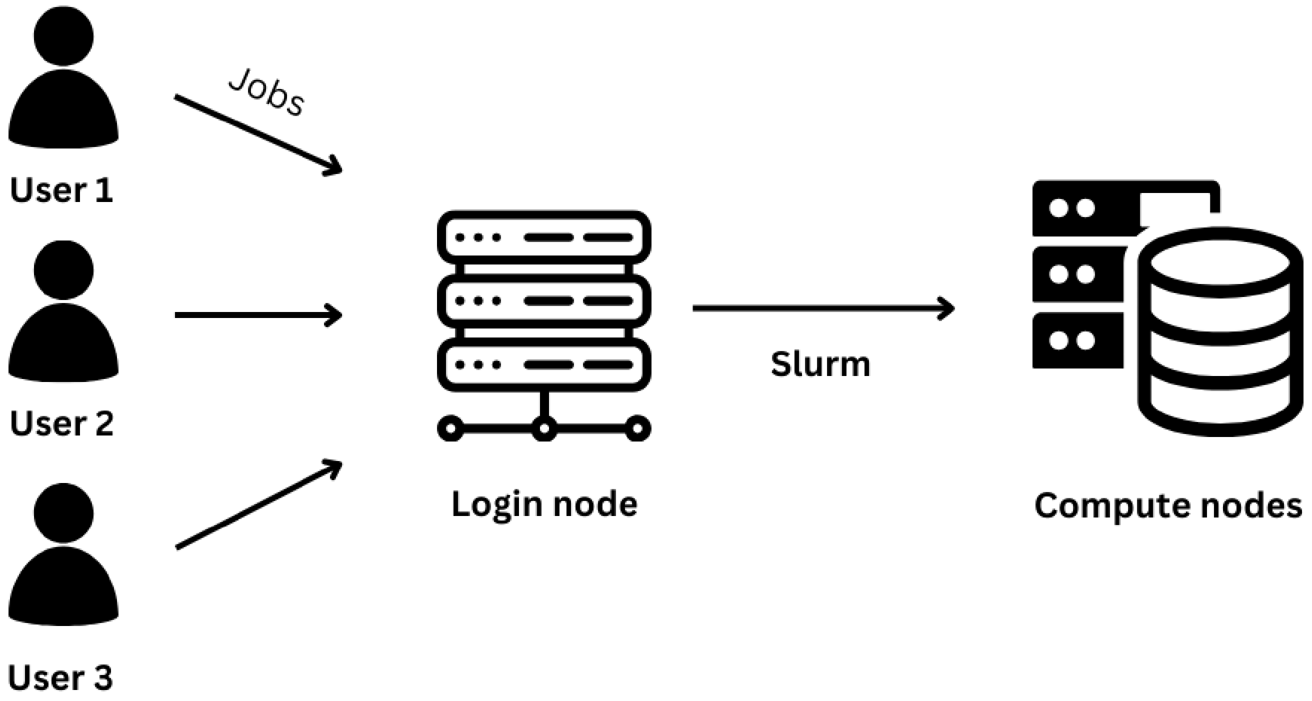
ICT Solutions for Brilliant Minds

Batch job systems

Running applications with supercomputers

Batch queue system

- Supercomputers have a huge number of users
  - How to ensure effectiveness and fair resource sharing?
- Queuing system needs to be in place
  - No instant access to computing; Only when there are resources available
  - CSC uses SLURM for this


User 1
Jobs
User 2
User 3
Login node
Slurm
Compute nodes

How do we describe our job to Slurm?

- User needs to create a job script describing their job's needs
  - What resources they will need
  - How much time the job should take
  - Which supercomputer "partition" it will use
- Based on this, the job might get accepted faster or slower
  - Typically, long jobs take more time to be accepted
  - Short and resource-light jobs are often accepted almost instantly

Example Slurm script

```
#!/bin/bash
#SBATCH --job-name=example
#SBATCH --account=<project_id>
#SBATCH --partition=debug
#SBATCH --time=00:10:00
#SBATCH --nodes=2
#SBATCH --ntasks-per-nodes=128

# srun launches "nodes * ntasks-per-nodes" copies of myprog
srun myprog
```

How do we send a job to the queue?

- Once you have your script ready you can:
- Submit your batch job script to the queue using "sbatch"

```
sbatch my_job.sh
```

- You can follow the status of your jobs with "squeue":

```
squeue –u $USER
```

- If something goes wrong, you can cancel your job with "scancel ":

```
scancel jobid                                (numeric ID of the job)
```

- Show job resource usage (for completed jobs) with sacct :

```
sacct jobid
```

Reservations in Slurm

- We have a reservation in Slurm today
- When running programs today, we'll add our reservation into our Slurm configuration
- It can be specified as the following in the slurm job file:

```
#SBATCH --reservation=scc-cpu
```

Send your own job to the queue

1. Log in to Mahti
2. Go to our Training folder and into "exercises"
3. Copy the example Slurm file into the "using_make" folder (with "cp")
4. Look at the Slurm example code and modify it (with e.g. Nano)
   a. Modify the resources so that you have 4 MPI processes reserved
   b. Change the last line to run your executable
5. Exit the editor (ctrl-x + ctrl-y) and send your job to the queue with sbatch
6. Type in "squeue –u $USER" to see if your job is waiting

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Thank you!

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